Prasad Sadanand Mahabare

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TECHNICAL SKILLS

- **Languages / Databases:** Embedded C, Linus, Python
- **Developer Tools:** VsCode, Keil, MPLabx, STM32CubeIDE
- **Simulation Platforms:** Proteus8, PicSimLab, VMWare
- **Protocols:** UART, I2C, SPI
- **Ticketing Systems:** ServiceNow, Footprints, TeamDynamix

PROFESSIONAL EXPERIENCE

Student Network Engineer, Purdue University **Jan 2023 – Dec 2023**

- Configured and installed Cisco catalysts (2650s to 9300s), firewalls, load balancers, DNS.
- Maintained Purdue Cloud System, performance by applying hot fixes and security configurations.
- Developed scripts to automate firmware upgrades over 800 network devices.

ServiceNow Associate, Purdue University Fort Wayne **Feb 2022 – Jan 2023**

- Automated tickets for 20 helpdesk associates based on time, rating, and expertise-based criteria.
- Agent auto assignment improved the help desk performance by 61% and customer satisfaction.
- Initiated Help Desk Training Module by analyzing daily challenges faced by current employees with knowledge base.

EDUCATION

Master of Science in Computer Science **Jan 2022 - Dec 2023**
Purdue University, Fort Wayne, Indiana

Bachelor of Engineering Computer Engineering **Aug 2016 - Aug 2019**
Pune University, Pune, India

PROJECTS

STM32 (ARM Cortex-M4)

- Learning ARM **Cortex-M4 architecture**, **core data flow**, and **execution modes** (privileged/unprivileged) for secure and efficient embedded system design.
- Developing judgment on when to use **STM32 HAL/LL libraries** versus **bare-metal programming** based on performance, control, and maintainability trade-offs.
- Interpreting and applying STM32 documentation **(reference manuals, datasheets, application notes)** for correct implementation of peripherals.

PIC Embedded System

- Built small applications using **GPIO**, **timers**, **serial communication**, and **interrupts**, emphasizing timing and signal generation.
- Integrated external components such as **LEDs**, **switches**, **LCDs**, and generated hex files using standard PIC development tools.

8051 Embedded System

- Gained hands-on experience with the 8051-microcontroller architecture using Assembly and Embedded C, including **instruction sets** and **addressing modes**.
- Implemented timer-based logic, **UART communication**, interrupt handling, and **I/O port control** to understand low-level firmware behavior.